

Persistence of trace metal contamination in a fluvial system: Lahn River, central Germany

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ABSTRACT

To monitor short-term changes in trace metal concentrations (Cu, Pb, Zn) at geomorphically sensitive floodplain locations and examine the persistence of contaminants in the fluvial environment, near-channel sites along the Lahn River, central Germany, that were sampled in 2007 were revisited in 2019. Thirty-three cores were extracted from sites <5 m from the active channel along a 75-km reach of the river. The median concentrations of Cu and Zn in 2019 were statistically unchanged from concentrations in 2007, but median concentrations of Pb in 2019 were statistically higher than in 2007. Concentration of all three trace metals remain above the background concentrations, but only Zn concentrations exceed the 2 σ background upper limit. On an individual sample site basis, the Cu concentration increased at 48 % of the sites and the Zn concentration increased at 45 % of the sites between 2007 and 2019. Pb concentration increased at 70 % of the sites. Concentrations of trace metals generally increased with distance downstream; the lowermost reach of the Lahn River continues to have the highest concentration of trace metals. Between 2007 and 2019, the concentrations (mean and median) of all three metals increased in the uppermost and lowermost sections of the study area whereas in the middle section the mean and median concentrations of Cu and Zn and the median concentration of Pb decreased (the mean concentration of Pb increased in the middle reach). Changes in metal concentrations between 2007 and 2019 do not appear to be caused by factors such as changes in substrate of samples, variable sedimentation rates and flood history, or the percentage of clay and organic matter content. Elevated concentrations of trace metals persist in near-channel sediments and will continue to pose a risk of future downstream heavy metal pollution in the study reach, especially along the lowermost reach of the Lahn River where the most contaminated sediments are located. Trace metal concentrations in the study reach appear to have become “uncoupled” from overall decreases in metal releases to the environment.

1. Introduction

Many river systems throughout Europe reflect the legacy of long-term exposure to pollutants, including trace metals. Often the product of industrial activity or mining, trace metals have in some locations been released to European rivers for centuries (Matschullat et al., 1997; Brännvall et al., 1999; Bindler et al., 2011; Kincey et al., 2018; Wang et al., 2019). The release of trace metals to the environment was at a maximum from the early- to mid-nineteenth century until the early 1970s and has decreased since then (Renberg et al., 2001; Middelkoop, 2002; Ciszewski and Grygar, 2016; Dhivert et al., 2016; Wang et al., 2019). Historical events in Europe since the late 1980s and growing awareness of the dangers of trace metals in the environment have resulted in cleaner rivers and floodplain sediments across much of western Europe (Brügmann, 1995; Middelkoop, 2002; Zerling et al.,

2006; Martin, 2015; Wang et al., 2019). Technological advances in water treatment plants and industrial facilities in eastern Europe further reduced the spread of air and water pollution in Europe (Scherer et al., 2003; Zerling et al., 2006).

In Germany, surface waters have become markedly cleaner over the last few decades (Wang et al., 2019), but high concentrations of trace metals remain along many rivers. Metals are often stored in channel banks and near-channel sediments, geomorphically sensitive areas where sediments and attached pollutants are easily eroded (James, 1989; Förstner, 2003; Foulds et al., 2014; Clement et al., 2017). Sediment samples collected shortly after major flood events have established that high concentrations of trace metals may be transported by high discharges and subsequently deposited in downstream floodplains (Förstner, 2003; Krüger et al., 2005; Schwartz et al., 2006; Foulds et al., 2014; Le Gall et al., 2018). Although the release of trace metals to the

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environment is less today than in the past, trace metals stored in floodplain sediments, named *legacy sediments* by James (2013), should be regarded as future sources of trace metal pollution (Schwartz et al., 2006; Ciszewski and Turner, 2009; Dennis et al., 2009; Foulds et al., 2014; Wadige et al., 2016; Esser et al., 2020).

Urbanized, industrial, or mined watersheds, which typically generate the highest releases of trace metals to the environment, have been the focus of numerous studies over the last two decades (e.g., Middelkoop, 2002; Macklin et al., 2006; Ciszewski and Turner, 2009; Dennis et al., 2009; Resongles et al., 2014; Ilie et al., 2017; Matache et al., 2018; Szabó et al., 2020). Much of this work documents anthropogenic enrichment of trace metals along rivers downstream from urbanized, industrial, or mined areas in which floodplain trace metal concentrations exceed background concentrations, or values that reflect non-anthropogenic metal concentrations (Birch et al., 2000; Matschullat et al., 2000; Bindler et al., 2011; Štrbac et al., 2018; Geiselbrecht et al., 2019). Less is known about trace metal storage in non-urbanized, non-industrial areas, or non-mined areas (an exception is Murad and Fischer, 1978).

The Lahn River basin is a relatively agrarian watershed with a drainage area of $\sim 5000 \text{ km}^2$. Trace metals are clustered in near-channel positions of the floodplain ($< 5 \text{ m}$ from active channel), with lower concentrations present with distance from the near-channel zone (Martin, 2009). The concentration of trace metals (Cu, Pb, Zn) along the Lahn River is less than that noted along waterways such as the Rhine or Elbe, although individual sites have metal concentrations approaching levels associated with urbanized, industrial, or mined watersheds (Martin, 2004). Trace metal concentrations are higher along the Dill River, a tributary to the lower Lahn River, and the lower Lahn River itself; both the Dill River and lower Lahn River have a history of upstream industrial activity and mining (Martin, 2015), although mining appears to be less important than industrial activity in contributing trace metals to the watersheds (Martin, 2019). Vertically, surface samples collected at numerous sites along the Lahn and Dill rivers generally contain lower concentrations of trace metals than do samples collected at greater depths (Martin, 2015), a reflection of continuing declines in metal releases to the environment that has been documented elsewhere (e.g., Middelkoop, 2002; Zerling et al., 2006; Le Gall et al., 2018; Buchty-Lemke et al., 2019).

The present study extends previous research on changes in heavy metal concentrations at channel bank locations along a 75-km reach of the Lahn River between Marburg and Weilburg, documented in Martin (2004, 2012), and explores the persistence of trace metals in the floodplain environment. Near-channel locations sampled in 2000 and 2007 were revisited and resampled in 2019 to evaluate the persistence of trace metals along the Lahn River. Specific objectives of the study are to (i) determine changes since 2007 in trace metal concentrations in the upper 5 cm of floodplain soils at near-channel ($< 5 \text{ m}$ from active channel) locations sampled in 2007, (ii) examine changes between 2007 and 2019 along the upper vs. lower reaches of the Lahn River, (iii) assess continued contributions of the Dill River to trace metals concentrations along the lower Lahn River, and (iv) assess the conveyance of trace metals during a flood through sampling of flood sediments deposited in March 2019 along a reach of the upper Lahn River. This study extends my work on short-term changes in the concentrations of heavy metals and allows evaluation of the persistence of metal storage in channel bank sediments of the Lahn River. Channel bank positions are relatively active geomorphically and regarded as potentially significant sources for future heavy metal pollution (James, 1989; Marcus, 1989; Leigh, 1997; Lecce and Pavlowsky, 2001). Finally, this work presents an opportunity to assess whether declines over the last several decades in trace metal releases to the environment are reflected in trace metal concentrations at near-channel positions along the Lahn River, or whether there is a lag in the response of metal concentrations in the floodplain to declines in metal releases to the environment.

2. Study area

Starting in the uplands north of Marburg, the Lahn River drains about 5000 km^2 of predominantly rural landscapes (Fig. 1). In the 75-km long study reach between Marburg and Weilburg, the Lahn valley is broad, reaching a maximum width of 2.5 km just downriver of Giessen. Urban centers of Marburg, Giessen, and Wetzlar, with most industrial activity concentrated in Wetzlar, contrast with the rural and agrarian land use elsewhere in the basin. Nearly three-quarters of land in the drainage basin is classified as agriculture with lesser amounts of forested land (16 %) and urban land (14 %), according to Jöckel (1981). Personal observation from nearly 25 years of working in the study reach indicate little or no land-use change along the floodplain and uplands over the study period. Uplands are typically forested and floodplains are cultivated or grazed. Channel bank riparian vegetation consists of scattered deciduous trees and a $\sim 3\text{-m}$ -wide near-channel vegetation zone composed of low-lying brush. Cultivation, where present, generally does not extend to the channel bank. Near-channel surface soils have a mean pH of 5.2 (standard deviation 0.5), a clayey-silt texture (60 % silt and clay, clay content 15–20 %), and organic matter content of ~ 10 %. Large-scale mining in the Lahn River watershed is not documented between Marburg and Wetzlar, but there is a history of iron mining during the late nineteenth century and much of the twentieth century in close proximity to the Lahn River between Wetzlar and Weilburg (Meyer, 2007). Iron ore was mined during the same period along the Dill River, which empties into the Lahn at Wetzlar. The last iron mine in the study area closed in the early 1980s.

The Lahn River was extensively modified through deepening and channelization beginning in the mid-nineteenth century. As part of the modifications, many small step-dams were built, which have the effect of reducing flow velocity and stream power. The channel gradient along the Lahn River is low, generally between 0.02° and 0.14° . Although it is generally a quiet, tranquil river, flooding and overbank flow do occur, typically between January and April. The high flows are generated by late winter and early spring precipitation and early spring snowmelt in upland areas. Overbank flows have the capacity to erode channel banks and deposit fine sediments on point bars and near-channel surfaces as was the case in March 2019 during the field work for this study. A series of aerial photographs taken between 1935 and 1995 and personal field experience in the study reach suggest that the Lahn River channel has been stable for at least several decades. Meander swales scattered across the floodplain in the study reach provide evidence that lateral channel migration has occurred in the study reach, likely prior to mid-1800s channelization.

3. Methods

Locating the 33 sites where channel bank sediments were collected in 2000 and 2007 involved using GPS coordinates contained in field notes, topographic maps, and ground-based photographs (Fig. 1). As was the case in 2000 and 2007, sample sites are positioned $< 5 \text{ m}$ from the active river channel. I used an Oakfield handheld corer at each sample site to collect a 2.5 cm diameter core from the floodplain. The upper 5 cm of each core was sampled and placed in sterile plastic Whirlpack bags for transport to the geomorphology lab at Justus Liebig-Universität, Giessen. As was done in 2007, the latitude and longitude of each sample site was recorded using a handheld GPS unit. Standard sediment analyses were done on one half of each sample in the geomorphology lab. Grain size analysis involved determination of the proportions of clay ($< 2 \mu\text{m}$) and silt ($2\text{--}63 \mu\text{m}$) using the pipette method and sand percentage through wet sieving. Organic matter content was measured by loss on ignition (LOI) at 430°C for 24 h (Davies, 1974).

The other half of each sample was submitted to ALS Chemex Labs (an ISO 9002 registered analytical laboratory in North America) for analysis of trace metals. Samples were submitted to the same ALS Chemex lab facility in North Vancouver, BC, Canada, that I have used in my previous

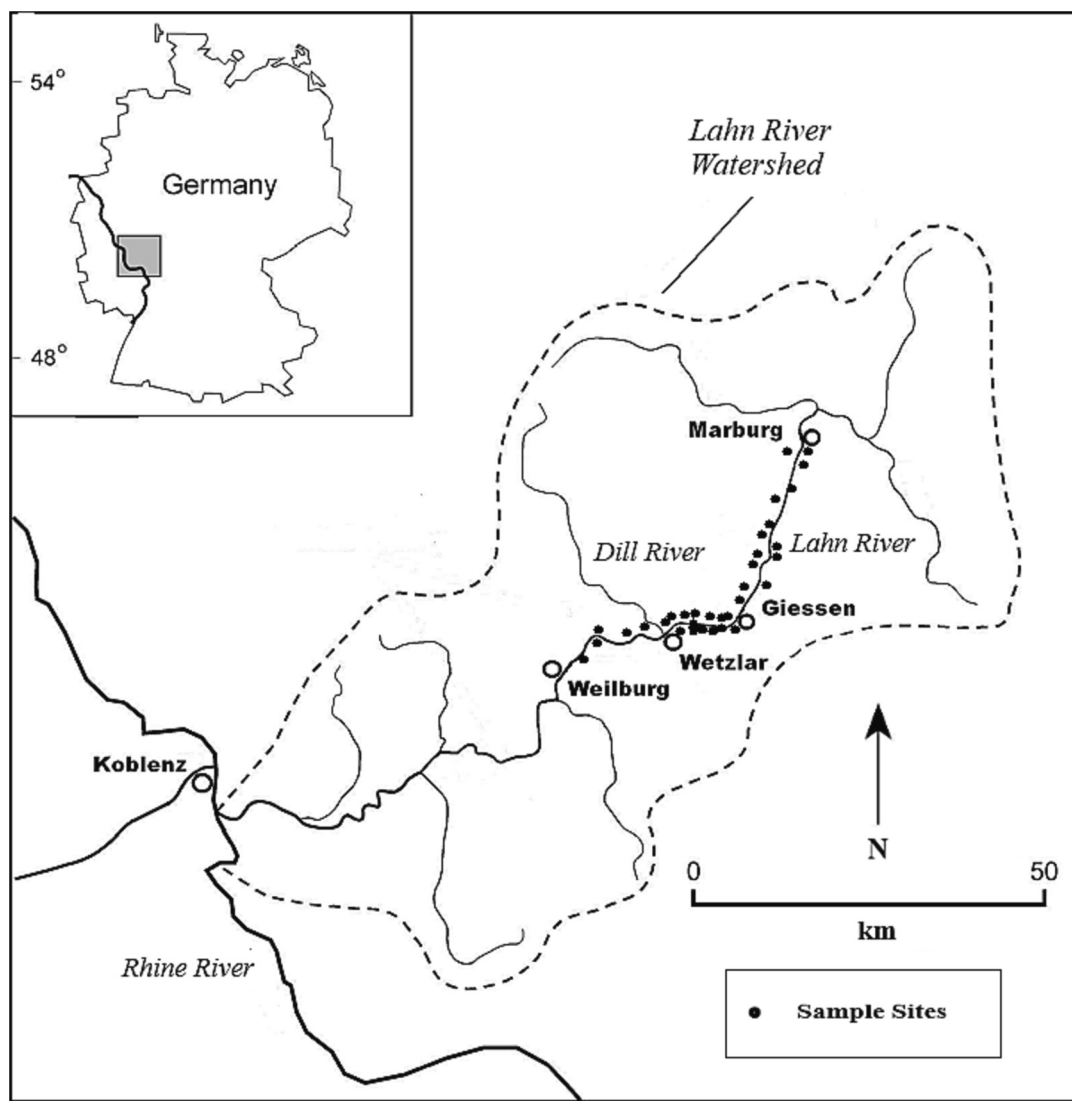


Fig. 1. Study reach of the Lahn River between Marburg and Weilburg, Germany. Sample sites (solid dots) represent locations where sediment cores were collected.

work along the Lahn River and identical laboratory preparation procedures were followed (air drying of samples, screening to 63 μm). The <63 μm fraction size from each sample was digested in perchloric, nitric, and hydrofluoric acids (triple-acid digestion). Based on work by [Lecce and Pavlowsky \(1997\)](#), this process yields near total recovery of metals. Inductively coupled plasma atomic emission spectroscopy (ICP-AES) was used to measure concentrations of Cu, Pb, and Zn, with an analytical sensitivity of 1.0 mg kg^{-1} for Cu and 2.0 mg kg^{-1} for Pb and Zn. Coefficients of variation on repeated measurements of standards, as supplied by quality control data from ALS Chemex, were 2.8 % for Cu, 3.5 % for Pb, and 2.6 % for Zn. Analysis of duplicates produced a mean difference of 1.1 mg kg^{-1} for Cu, 2.1 mg kg^{-1} for Pb, and 8.8 mg kg^{-1} for Zn.

To control for substrate influence on metal concentrations, concentrations were normalized by dividing them by Al and Fe concentrations and clay and organic matter (OM) content (all reported in %). Normalizing to “conservative” elements such as Al and Fe is a commonly used technique to detect anthropogenic metal contributions. Early descriptions of this technique are found in [Salomons and Förstner \(1984\)](#), [Schropp et al. \(1990\)](#), and [Horowitz \(1991\)](#), and it continues to be used in a variety of geomorphic settings (e.g., [Loring, 1991](#); [Ho et al., 2012](#)). Grain size and organic matter content have also been shown to influence metal concentrations ([Horowitz, 1991](#); [Owens et al., 2005](#)), whereas

others find no correlation between metal concentrations and organic matter or fine sediment contents ([Lokas et al., 2010](#)). Near-channel alluvium along the Lahn River typically has a clayey-silt texture (~60 % clay and silt), with clay contents around 15–20 %. The organic matter content of surface sediments is typically 10 % ([Martin, 2009](#)).

Thirty-five background samples, which are assumed to represent the pre-industrial age (pre-1750 A.D.) metal concentrations, were collected at various locations along the study reach of the Lahn River at depths of 0.85 to 1.15 m. They serve as the basis for measuring the magnitude of anthropogenic heavy metal enrichment in surface soils. Background samples were collected at channel bank sites where surface samples were collected ($n = 18$) and at sites in the study reach that were < 10 m from the Lahn River channel ($n = 17$). Background samples could not be collected at all surface sample sites because gravel layers or a high water table prevented coring to depths sufficient to reach pre-industrial age sediments. The trace metal concentrations in background samples were determined by ALS Chemex (North Vancouver facility) with the same procedures used for analyzing surface samples. The determination of natural background, or nonanthropogenic, metal concentrations has generated considerable discussion in the literature (e.g., [Birch et al., 2000](#); [Matschullat et al., 2000](#); [Bindler et al., 2011](#); [Geiselbrecht et al., 2019](#)). To obtain the likely upper limit on background values and better assess the degree of anthropogenic metal contamination, I used the 26

method of Matschullat et al. (2000). As outlined in my earlier work (Martin, 2004), I use deeply buried sediments to establish natural background metal concentrations in this study. Evidence that deeply-buried sediments are appropriate for establishing background concentrations comes from several AMS conventional radiocarbon ages on charcoal recovered from depths >0.60 m along the Dill River (Martin, 2015). The radiocarbon ages indicate that sediments at those depths are pre-nineteenth century in age.

In addition to comparing sample metal concentrations to those of background values, several published standards provide a means to evaluate the degree of soil contamination. Standards for assessing the degree of soil contamination in Germany are outlined in the German Federal Soil Protection Act of 1998 (*Bundes-Bodenschutzgesetz {BbodSchG}*) and the German Federal Soil Protection and Contaminated Sites Ordinance of 1999 (*Bundes-Bodenschutzverordnung {BbodSchV}*). Precautionary values (mg kg⁻¹ for dry weight fine soil), the value at which natural soil functions are said to be at risk, are outlined in the protection act and ordinance. Here, the metal concentrations from the 2000, 2007, and 2019 samples were compared to the precautionary values as an additional measure of the degree of soil contamination. Finally, the threshold effect concentration (TEC) and probable effect concentration (PEC) guidelines developed by MacDonald et al. (2000) were compared to the 2000, 2007, and 2019 metal concentrations. Adverse effects are deemed “unlikely” when metal concentrations fall below the TEC. Concentrations above the PEC will likely have negative effects on the soil.

4. Results

4.1. Mean metal concentrations, 2007 and 2019

The near-channel median concentrations of Cu and Zn were statistically unchanged, whereas the concentration of Pb increased, between 2007 and 2019 (Table 1). Based on a Mann-Whitney *U* test comparison of median values, the increase in Pb concentration was statistically significant ($p = 0.03$). Interestingly, the maximum values as reflected in the ranges for all three metals were greater in 2019 than in 2007, with the increase most dramatic for Pb (65 mg kg⁻¹ in 2007 compared to 170 mg kg⁻¹ in 2019). Median concentrations of Cu, Pb, and Zn are all statistically greater ($p \leq 0.001$) than the “background” or non-

anthropogenic concentrations from 35 sites in the study area. When compared to the 26 upper limit for the background samples (Matschullat et al., 2000), the mean values of Cu and Pb in 2019 are below the upper limit whereas the mean value of Zn is above the upper limit. Concentrations of all three trace metals are less than the PEC (probable effect concentrations) of MacDonald et al. (2000), defined as concentrations above which adverse effects to the environment are likely. Concentrations are greater than the TEC (threshold effect concentration) of MacDonald et al. (2000) for all three trace metals. The TEC represents the level below which adverse environmental effects are unlikely. Concentrations of Cu and Pb are roughly equal to (Cu) or less than (Pb) the “Precautionary Values” of the German Federal Soil Protection Act of 1998, whereas Zn concentrations are greater than the “Precautionary Value”. Natural soil functions are said to be at risk when the “Precautionary Value” is exceeded (*Bundesbodenschutzgesetz {BbodSchG}*).

Beyond the summary values, comparison of concentrations at individual sample sites with the environmental standards reveals spatial variability in trace metal contamination in the study reach between the two sample dates. In 2007, 14 of 33 sites exceeded or equaled the precautionary value of 40 mg kg⁻¹ for Cu; by 2019, 15 sites exceeded or equaled it. No sample sites in 2007 or 2019 had Cu concentrations that exceeded the PEC value (197 mg kg⁻¹), but 21 sites in 2019 and 23 sites in 2007 exceeded the TEC value (35.7 mg kg⁻¹). No sites in 2007 had Pb concentrations that exceeded or equaled the precautionary value (70 mg kg⁻¹); by 2019, four sites exceeded or equaled the value. In 2007, no sample sites had Pb concentrations exceeding the PEC (91.3 mg kg⁻¹), but 31 sites had concentrations exceeding the TEC (35 mg kg⁻¹). By 2019, one sample site exceeded the PEC and 31 sites exceeded the TEC. For Zn, the precautionary value of 150 mg kg⁻¹ was exceeded or equaled at 32 sites in 2007 and at 31 sites in 2019. The PEC for Zn (315 mg kg⁻¹) was exceeded at five sample sites in both 2007 and 2019. The TEC (123 mg kg⁻¹) for Zn was exceeded at all sites in 2007 and at all but one site in 2019. On a sample site basis, the Cu concentration decreased at 16 sites (48 %) between 2007 and 2019, increased at 16 sites (48 %), and remained unchanged at one site (4 %). At 23 sites (70 %) the Pb concentration increased between 2007 and 2019; it decreased at the other 10 sites (30 %). The Zn concentration in 2019 was higher at 15 sites (45 %) and lower at 18 sites (55 %).

4.2. Spatial pattern of metal concentrations

Examining the spatial pattern of changes in metal concentrations between 2007 and 2019 reveals that changes vary for different reaches of the study area. As was the case in 2000 and 2007 (Martin, 2012), trace metals in 2019 increased in a downstream direction (Fig. 2), but site-to-site variability for each trace metal continues to be evident. To examine the spatial variability of each trace metal across the sample dates, I divided the study area into three reaches in a downstream direction: Marburg to Giessen (Mb-Gi), Giessen to Wetzlar (Gi-Wz), and Wetzlar to Weilburg (Wz-Wb). For Cu and Zn, the variability is greatest in the Gi-Wz section; in contrast, for Pb the site-to-site variability differs little across the three sections. Table 2 provides summary descriptive statistics for the three trace metals in 2007 and 2019 for the three river sections. The median Cu, Pb, and Zn concentration increased at two of the three river sections (Mb-Gi and Wz-Wb) between 2007 and 2019; the median concentrations of all three metals decreased between 2007 and 2019 in the Gi-Wz river reach. Using the 26 upper limit for the background samples as a conservative measure of anthropogenic metal enrichment, mean Zn concentrations were greater than the 26 limit in all three reaches (2007, 2019), mean Cu concentrations were greater than the 26 limit in the Wz-Wb reach only (2007, 2019), and the mean Pb concentrations were within the 26 limit in all three reaches (2007, 2019). The Wz-Wb segment continues to have in 2019 the highest median and mean concentrations of all three trace metals, just as it did in 2000 and in 2007 (Martin, 2012).

The higher metal concentrations downstream of Wetzlar likely

Table 1

Summary descriptive statistics for soil samples; median concentrations shown in bold are statistically different ($p = 0.05$) between 2007 and 2019 samples; standard deviations (σ) shown in parentheses; the “Precautionary Value” is published in the German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz*) of 1998; the TEC (threshold effect concentration) and PEC (probable effect concentration) values are from MacDonald et al. (2000); 26 background upper limits (Matschullat et al., 2000); all values are for dry weight sediment.

	Metal concentrations (mg kg ⁻¹)		
	Cu	Pb	Zn
2019 ($n = 33$)			
Median	39	54	228
Mean (σ)	40.2 (10.9)	58.4 (24.6)	239.9 (66.7)
Range	22–61	27–170	111–384
2007 ($n = 33$)			
Median	38	46	223
Mean (σ)	41.1 (8.9)	48.4 (9.9)	243.3 (57.1)
Range	27–56	34–65	146–343
Precautionary Value	40.0	70.0	150.0
TEC	35.7	35.0	123.0
PEC	197.0	91.3	315.0
Background ($n = 35$)			
Median	20	20	72
Mean (σ)	22.9 (12.7)	26.6 (23.3)	84.1 (43.8)
26 Upper Limit	48.3	73.2	171.7
Range	11–82	12–136	42–246

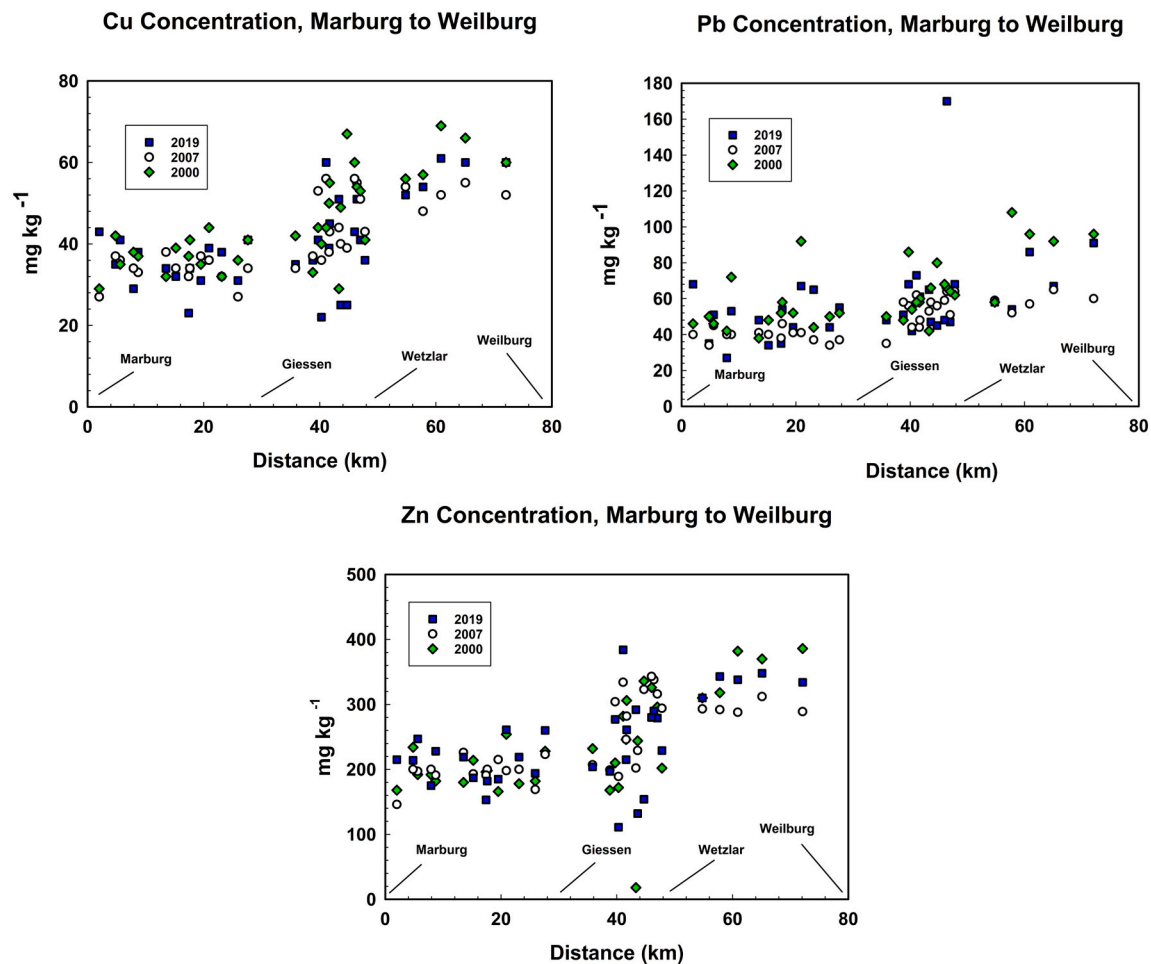


Fig. 2. Downstream changes in Cu, Pb, and Zn concentrations, 2000, 2007, and 2019. Locations of Marburg, Giessen, Wetzlar, and Weilburg are noted on the horizontal axis. The confluence of the Dill River and the Lahn River is at Wetzlar.

reflect the input of metal-contaminated sediments from the Dill River, which empties into the Lahn River in Wetzlar. The Dill River and Wz-Wb reach of the Lahn River drain areas that were extensively mined for iron from the late nineteenth century until the early 1980s. [Martin \(2015\)](#) documented that high concentrations of Cu, Pb, and Zn are stored in geomorphically-sensitive, near-channel locations along the Dill River, where they may be periodically conveyed downstream to the Lahn River. The source of the metals appears to be upstream industrial facilities in the Dill River, not the historic iron mining activity that occurred in the basin until the late twentieth century ([Martin, 2019](#)).

The 2019 data indicate that trace metal concentrations, although increasing overall in a downstream direction, are spatially variable from site to site. “Hotspots” with anomalously high concentrations may exist at one location, but immediately downstream at the next sample location the concentrations are much lower. This complex, irregular spatial pattern was also apparent in the 2000 and 2007 samples and has been noted in other studies of trace metals in near-channel positions (e.g., [Coulthard and Macklin, 2003](#); [Lecce and Pavlowsky, 2014](#)). It is usually attributed to geomorphic processes that control storage and transport of metal-contaminated sediments ([Dhivert et al., 2016](#)), differences in topographic relief ([Lecce and Pavlowsky, 2001](#); [Middelkoop, 2002](#)), or the presence of multiple sources of trace metals along a study reach ([Lecce and Pavlowsky, 2014](#)). In the study reach, there is little change in topographic relief in near-channel positions. Multiple potential sources of trace metals are found along the study reach, although the number of large point sources on or close to the floodplain (e.g., large industrial facilities) is limited to one such facility in the Mb-Gi section; samples

collected from two locations immediately downstream (<1 km) from that facility have Cu (39, 38 mg kg⁻¹), Pb (67, 65 mg kg⁻¹), and Zn (261, 219 mg kg⁻¹) concentrations that are slightly higher than concentrations at two sites immediately upstream (Cu: 31, 34 mg kg⁻¹; Pb: 44, 54 mg kg⁻¹; and Zn: 185, 182 mg kg⁻¹). Downstream from Giessen there are more industrial facilities as well as the input of trace metals from the Dill River that joins the Lahn River in Wetzlar.

Flooding along the study reach in March 2019 provided the opportunity to collect freshly-deposited sediment samples at a site near Giessen. These samples potentially provide an indication of the present conveyance of trace metals through the study reach during an episodic event ([Lecce et al., 2008](#)). Three samples were collected on 22 March 2019 with a trowel from silty sediments that were partially burying living grass within 5 m of the Lahn River. As indicated by the partial burial of grasses, the sediment was deposited by an overbank event that occurred a week earlier. The range of trace metal concentrations for the three freshly-deposited sediment samples (Cu: 34–40 mg kg⁻¹; Pb: 37–40 mg kg⁻¹; Zn: 213–248 mg kg⁻¹) are within the range for those metals in the study reach, although Pb concentrations from the samples fall in the lower part of the range. It is worth noting that the site where sediment samples were collected is at the lower end of the Mb-Gi section, the section with the lowest metal concentrations in the study area. Comparing metal concentrations in the freshly-deposited samples with those shown in [Table 2](#) indicates that concentrations of Cu and Zn in the freshly-deposited sediment samples are similar to the mean and median concentrations for those metals in the Marburg-Giessen river reach; the concentrations of Pb in the freshly-deposited sediment samples are less

Table 2

Summary descriptive statistics for soil samples by river segment. For a given metal and river segment, median concentrations shown in bold are statistically different ($p = 0.05$) between 2007 and 2019; standard deviations shown in parentheses. River segments are: Marburg–Giessen (Mb–Gi), Giessen–Wetzlar (Gi–Wz), and Wetzlar–Weilburg (Wz–Wb) (Figs. 1 and 2 show river segment locations).

	Metal concentrations (mg kg ⁻¹)		
	Cu	Pb	Zn
2019			
Mb–Gi ($n = 14$)			
Mean (σ)	34.9 (5.5)	48.6 (12.9)	209.9 (32.4)
Median	34.5	49.5	214.5
Gi–Wz ($n = 14$)			
Mean (σ)	39.3 (10.7)	63.7 (32.2)	236.1 (73.5)
Median	40	55	245
Wz–Wb ($n = 5$)			
Mean (σ)	57.4 (4.1)	71.2 (16.6)	334.6 (14.7)
Median	60	67	338
2007			
Mb–Gi ($n = 14$)			
Mean (σ)	33.6 (3.3)	39.8 (3.6)	195.4 (19.0)
Median	34	40	199
Gi–Wz ($n = 14$)			
Mean (σ)	44.6 (7.9)	53.7 (8.5)	271.9 (57.7)
Median	43	56	288
Wz–Wb ($n = 5$)			
Mean (σ)	52.2 (2.7)	58.6 (4.7)	294.8 (9.8)
Median	52	59	292
Background ($n = 35$)			
Mean (σ)	22.9 (12.7)	26.6 (23.3)	84.1 (43.8)
Median	20	20	72
2 σ Upper Limit	48.3	73.2	171.7

than the mean and median concentrations in the Mb–Gi river section. The metal concentrations in the three freshly-deposited sediment samples suggest that higher than background concentrations of the three metals continue to be transported through the upper Lahn River, probably eroded from upstream channel banks (Lecce and Pavlowsky, 2014).

4.3. Normalizing metal concentrations

To evaluate the potential influence of substrate on metal concentrations, metal concentrations were normalized for Al, Fe, clay and OM (all reported in %). A general look at the differences in these variables between 2007 and 2019 is provided in Table 3. Median clay and OM values between the two dates are not statistically different. Median values of Al and Fe were statistically lower in 2019 than in 2007, suggesting a potential difference in substrate between the 2019 and 2007 samples. The normalized concentrations of Cu, Pb, and Zn with Al and Fe for 2000, 2007 and 2019 are shown in Fig. 3. Comparison of the normalized concentration trends with the original concentration trends (Fig. 2) reveals only minor changes at a handful of sites. At a few sites

Table 3

Summary descriptive statistics for sample characteristics. Median concentrations shown in bold are statistically different ($p = 0.05$) between 2007 and 2019. Standard deviation represented by “ σ ”. “OM” represents organic matter content; “Cl” represents clay content. All values are in “%”.

	OM	Cl	Al	Fe
2019				
n	33	33	33	33
Mean	9.6	15.4	5.1	3.2
σ	3.2	4.7	0.7	0.7
Median	9.4	15.9	5.2	3.1
2007				
n	33	33	33	33
Mean	9.8	16.7	5.5	3.3
σ	2.5	3.2	0.3	0.2
Median	9.7	16.4	5.5	3.3

normalization with Fe or Al brings the values for the three years closer together, but at other sites normalization pulls the values apart. Overall, it appears that Al and Fe concentrations have little impact on metal concentration trends, suggesting that changes are a function of metal sources rather than of substrate.

As an additional normalization of trace metal concentrations, I compared metal ratio median values for the 2007 and 2019 samples to gauge the impact of substrate on median metal concentrations for the two periods (Table 4) and determine whether changes in metal concentrations are the result of increased metal input to the study reach or the result of changes in substrate. The three trace metals were normalized for Al and Fe. The difference in mean and median ratio values for Cu and Zn for the two periods is small and statistically insignificant. In contrast, the mean and median ratio values for Pb are larger in 2019 than in 2007, and the difference in the median ratio value is statistically higher in 2019 than in 2007. This suggests that the increase in Pb concentration over that period is a function of a higher input of Pb to the system rather than of changes in substrate.

To gauge the impact of Al, Fe, clay, and organic matter more closely on the 2019 samples, I isolated the five sites in 2019 with the highest Cu, Pb, and Zn concentrations and the five sites with the lowest Cu, Pb, and Zn concentrations, then compared their Al, Fe, clay and OM values (Table 5). The maximum median values of Cu, Pb, and Zn are statistically different than the minimum median values at the sites, but there is no statistical difference in the median Fe (%), clay (%) and OM content (%) between the maximum and minimum sites. In contrast, the median Al (%) is statistically higher in the maximum sites than in the minimum sites.

A series of bivariate regression plots (Fig. 4) for the 2019 data depict the relationship between metal concentrations (Cu, Pb, and Zn) and Fe, Al, clay, and OM content (all reported in %). Cu concentrations have a weak, but statistically significant, relationship to Al ($r^2 = 0.23$) and clay ($r^2 = 0.13$), but are not related to Al or OM content. Zn concentrations also have a weak, but statistically significant, relationship to Al ($r^2 = 0.26$) and clay ($r^2 = 0.14$), but are not related to Al or OM content. In contrast, the concentration of Pb is not statistically related to Fe, Al, clay, or OM content. It appears that the concentrations of the three trace metals are little affected by Al, Fe, percentage clay, and OM content.

5. Discussion

Returning to near-channel sites along the Lahn River that were sampled in 2000 and 2007, this study extends the record of change in trace metal concentrations to 2019 and provides an indication of the persistence of trace metal contamination along the Lahn River. For the 75-km study reach, this study reveals no statistical change in Cu and Zn concentrations, but an increase in Pb concentrations. Although median concentrations of all three trace metals are statistically greater than median background concentrations in both 2007 and 2019, only Zn exceeds the 2 σ upper limit for background concentrations. This suggests that the study reach is not heavily contaminated with anthropogenic trace metals, not unexpected given the relatively agrarian character of the Lahn River drainage basin. Further evidence of the degree of anthropogenic trace metal concentration in the study reach comes from comparing the 2019 trace metal concentrations to other sediment quality guidelines such as the TEC and PEC of MacDonald et al. (2000) and the “Precautionary Value” of the German Federal Soil Protection Act of 1998. Such comparisons show that the study reach is not highly contaminated, but also reveal that Pb contamination worsened slightly between 2007 and 2019 whereas Cu and Zn contamination was virtually unchanged.

The increased near-channel concentrations of Pb between 2007 and 2019 noted in this study may result from the availability of the metal in the Lahn River basin and periodic conveyance of it to the river channel and near-channel positions. Previous work along five transects perpendicular to the Lahn River (Martin, 2009) revealed that Pb and Zn

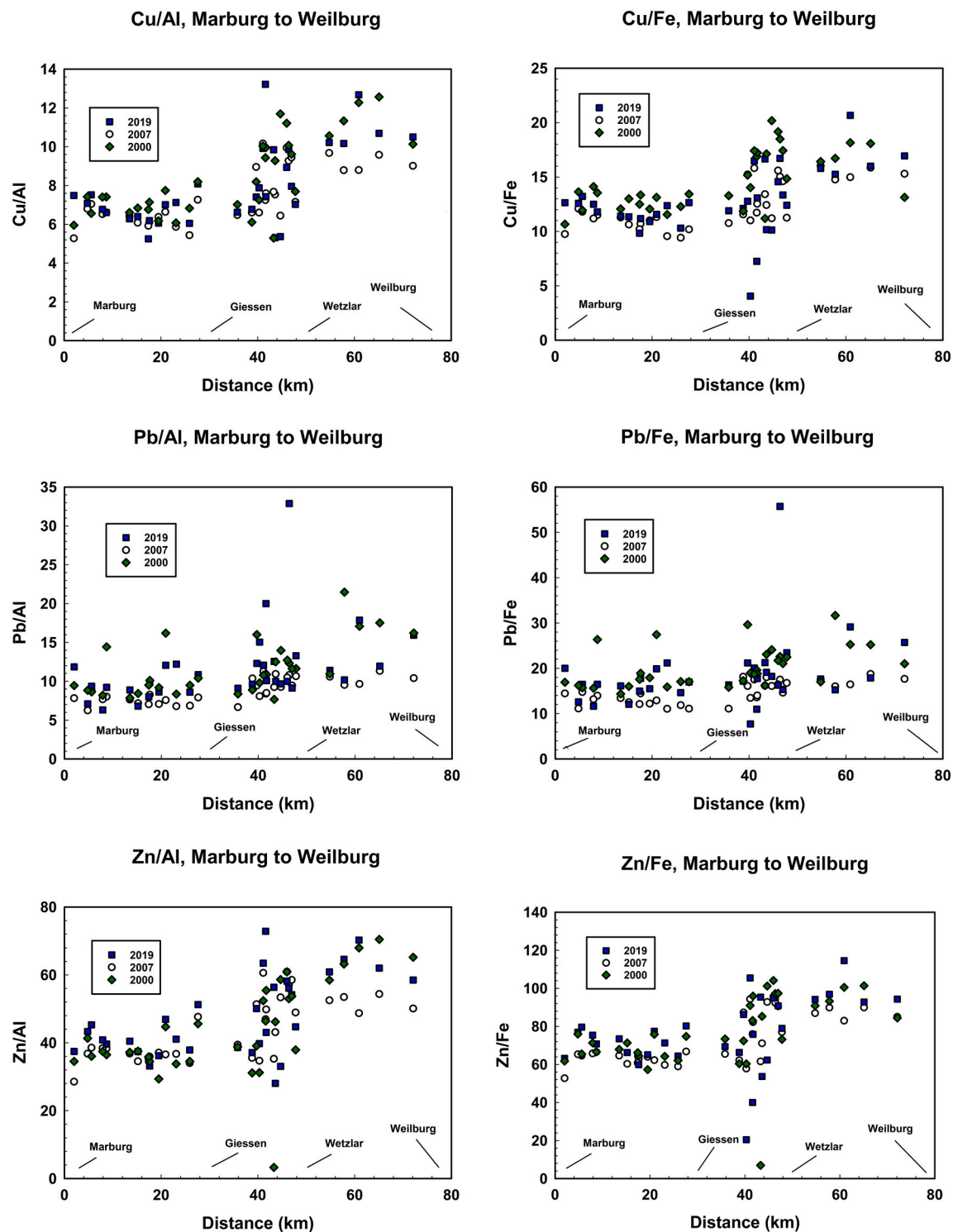


Fig. 3. Normalized downstream changes in Cu, Pb, and Zn concentrations, 2000, 2007, and 2019. Locations of Marburg, Giessen, Wetzlar, and Weilburg are noted on the horizontal axis. The confluence of the Dill River and the Lahn River is at Wetzlar.

concentrations in near channel locations and across the transects increased between 1995 and 2007 whereas Cu concentrations were unchanged or declined slightly. Across the transects in both 1995 and 2007, Cu and Zn concentrations were markedly higher immediately adjacent to the channel, then declined rapidly away from the channel and remained uniformly low across the floodplain. In contrast, Pb concentrations declined only slightly with distance from the channel and remained unchanged across the floodplain. These findings suggest that

spatial variability in Pb concentrations is limited laterally across the floodplain, whereas downstream near-channel spatial variability in Pb concentrations, as documented here, is relatively high. That is, large site- to-site Pb variability in the study reach is not present laterally across the floodplain, but is apparent downstream in near-channel locations, suggesting changes in the input of Pb to the near-channel environment over the study period. One potential source for additional Pb inputs to the river is the relatively higher concentrations of Pb

Table 4

Summary statistics of trace metal ratio values. Median concentrations shown in bold are statistically different ($p = 0.05$). Standard deviation represented by “ σ ”.

	Cu/Fe	Cu/Al	Pb/Fe	Pb/Al	Zn/Fe	Zn/Al
2019						
<i>n</i>	33	33	33	33	33	33
Mean	12.7	7.9	18.6	11.6	76.3	47.2
σ	3.1	2.0	7.9	4.8	18.8	11.8
Median	12.5	7.4	17.2	10.1	75.9	43.3
2007						
<i>n</i>	33	33	33	33	33	33
Mean	12.4	7.4	14.7	8.8	73.2	43.9
σ	2.1	1.4	2.4	1.5	13.2	9.1
Median	11.6	7.0	14.5	8.5	66.5	39.4

Table 5

Summary statistics sediment characteristics, maximum and minimum metal concentration, 2019 data. Maximum and minimum median concentrations shown in bold are statistically different ($p = 0.05$). Metal concentrations in mg/kg⁻¹; Fe, Al, organic matter (OM), and clay (Cl) in “%”. Standard deviation represented by “ σ ”.

	<i>n</i>	Mean	σ	Median
Cu (Max)	5	59	2.8	60
Al	5	5.5	0.5	5.6
Fe	5	3.5	0.3	3.5
OM	5	10.2	2	10.6
Cl	5	17.7	2.9	16.5
Cu (Min)	5	24.8	2.7	25
Al	5	4.2	0.8	4.4
Fe	5	3.4	1.4	2.5
OM	5	8.9	3.9	6.8
Cl	5	10.6	8	9.2
Pb (Max)	6	75.7	10.2	70.5
Al	6	5.5	0.4	5.6
Fe	6	3.3	0.3	3.3
OM	6	9.2	1.9	10
Cl	6	16.5	3.7	15.2
Pb (Min)	5	34.6	5.3	35
Al	5	4.3	0.9	4.4
Fe	5	3.1	1.3	2.8
OM	5	11.8	4.8	14.4
Cl	5	14.4	8.5	18.1
Zn (Max)	5	349.4	20	343
Al	5	5.5	0.5	5.6
Fe	5	3.5	0.3	3.5
OM	5	11.5	3.6	10.6
Cl	5	17.6	2.9	16.5
Zn (Min)	5	145	24.3	153
Al	5	4.2	0.8	4.4
Fe	5	3	1.4	2.5
OM	5	8.9	3.9	6.8
Cl	5	10.6	8	9.2

present in soils and surface sediments across the floodplain.

In addition to spatial variability in metal concentrations, temporal variability is captured in vertical metals trends from near-channel cores collected along the lower Lahn River (Martin, 2019) and Dill River (Martin, 2015). They revealed that metal concentrations vary with depth and appear to reflect peak releases of metals to the environment from upstream industrial activity, mining, or both. Although site-to-site differences exist, the vertical metal trends in cores are similar. This suggests that vertical metal trends reflect changing metal inputs to the river rather than inherent variability in metal concentrations. At least over the short term covered in this study and my previous work stretching back to 2000, temporal declines in trace metal concentrations to European watersheds (e.g., Grosbois et al., 2006; Meybeck et al., 2007; Le Gall et al., 2018; Wang et al., 2019) are not evident along this reach of the Lahn River. To the extent they represent the current geochemical load moving through the active channel (Lecce et al., 2008), the three “grab samples” collected from freshly-deposited

sediments at a site near Giessen shortly after a flood in March 2019 confirm that contaminated sediments are still being transported through the upper Lahn River. All three samples contain Cu, Pb, and Zn concentrations that exceed background levels.

Metal concentrations in 2019 generally increase with distance downstream, with maximum concentrations present downstream from Wetzlar. The Dill River, which empties into the Lahn River at Wetzlar, is likely the source of the higher metal concentrations as near-channel positions along the Dill River store relatively high concentrations of trace metals (Martin, 2015) that may be periodically conveyed to the Lahn River. Concealed within the general increase in metal concentrations downstream, there is variability among river reaches that I highlight by dividing the study reach into three river segments: Mb–Gi, Gi–Wz, and Wz–Wb. Mean Zn concentrations were greater than the 2 σ upper limit for background samples in all three segments in 2007 and 2019, mean Cu concentrations exceed the 2 σ upper limit in one segment (Wz–Wb) in 2007 and 2019, and mean Pb concentrations were less than the upper limit in all three segments in 2007 and 2019. Using this indicator, anthropogenic contamination with Zn is present along the entire study area, but contamination with Cu and Pb is less widespread. Mean concentrations of all three metals generally exceed the TEC baseline of MacDonald et al. (2000) in all three segments, but fall below the PEC baseline (the exception is Zn (2019) which is above the PEC value in the Wz–Wb reach. In summary, comparing the 2019 concentrations of Cu, Pb, and Zn with various background and baseline indicators reveals more anthropogenic contamination with Zn than with Cu and Pb, but evidence of at least minor anthropogenic storage of all three metals in near-channel positions, especially in the Wz–Wb river segment, is present.

The mean and median Cu, Pb, and Zn concentrations increased at two of three river segments between 2007 and 2019 (Mb–Gi, Wz–Wb). For the third river segment (Gi–Wz), the mean and median concentrations of Cu and Zn and the median concentration of Pb decreased between 2007 and 2019; the mean concentration of Pb increased in this river segment between 2007 and 2019. To ascertain whether observed changes in metal concentrations between 2007 and 2019 were the result of changes in metal conveyance to near-channel positions or some other factor (e.g., changes in substrate, differences in sediment ages for samples, localized flooding/deposition), metal concentrations in 2007 and 2009 were normalized by dividing them by Al, Fe, percentage clay, and organic matter content. The resulting ratios for the two dates were compared and plotted by distance for the 75-km study reach. In addition, characteristics of the 2007 and 2019 samples were compared by looking at the Al, Fe, clay, and organic matter contents for the two time periods.

Comparing the normalized metal downstream trends with the non-normalized metal trends reveals little change between the trends (Figs. 2 and 3). Concentrations of Cu, Pb, and Zn appear little affected by Al and Fe, suggesting that observed differences spatially are not the result of differences in substrate. Comparing the temporal changes in the metal ratios between 2007 and 2019 also suggests that differences in substrate are not an important factor. The mean and median ratios for Cu and Zn are statistically unchanged from 2007 to 2019, whereas the mean and median ratios for Pb increased from 2007 to 2019 (the difference in the median ratios is statistically significant). The Pb increase appears to be independent of a change in substrate. The values of organic matter content and percentage clay are not statistically different between the 2007 and 2019 samples; median values of Al and Fe are, however, statistically lower in 2019 than in 2007. Looking at the maximum and minimum Cu, Pb, and Zn concentrations for the 2019 samples only ($n = 5$ for each) reveals no statistical differences in Fe, clay, and organic matter content; the median Al concentration, however, is higher in the maximum concentration samples than in the minimum concentration samples. In aggregate, there is no compelling and consistent evidence that the concentrations of Cu, Pb, and Zn are strongly influenced by differences in substrate or the clay and organic

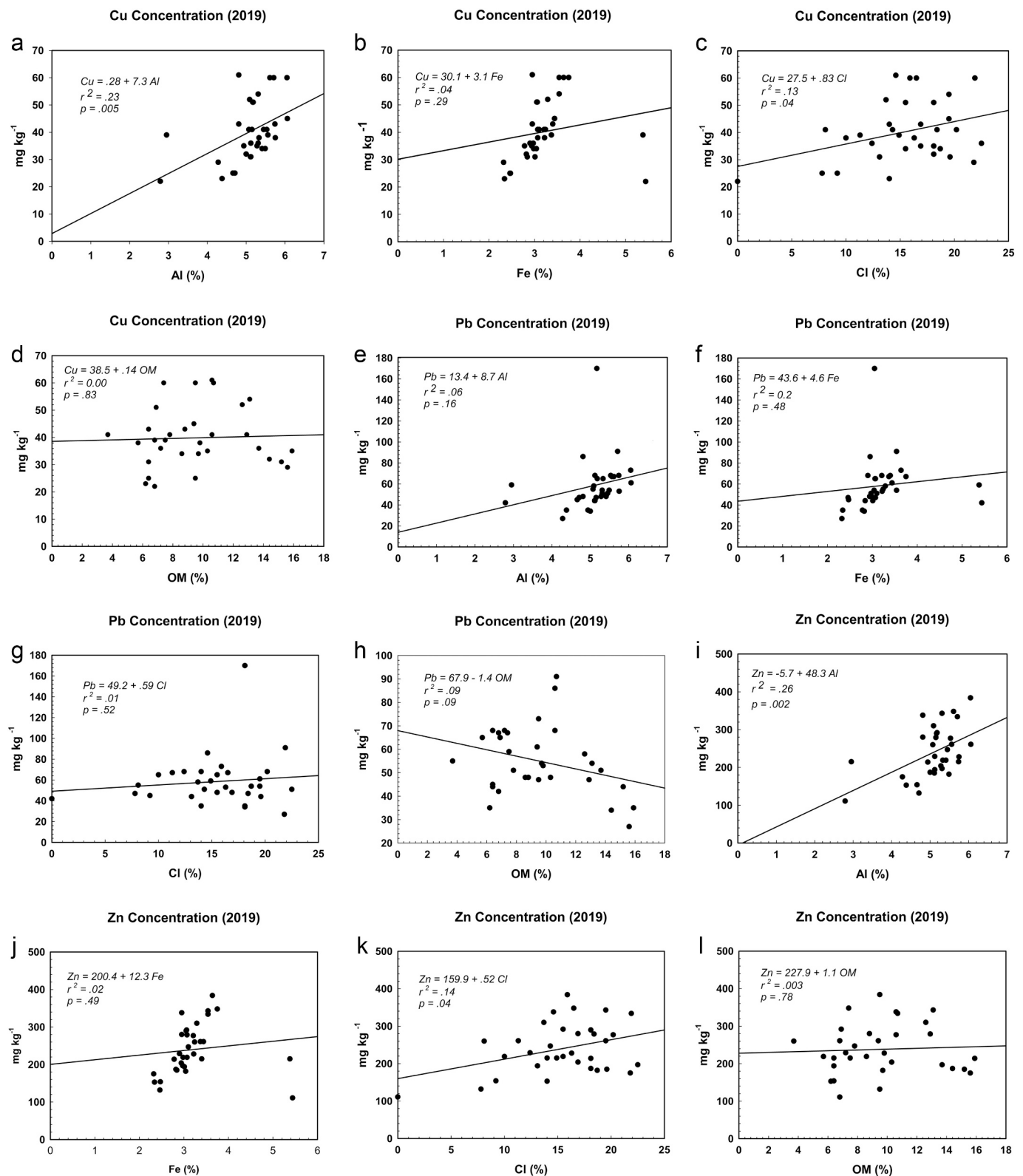


Fig. 4. Scatter plots and best-fit bivariate regression of metal concentrations and Al, Fe, percent clay (Cl) and percent organic matter (OM) for 2019 samples.

matter content. This last point is further supported by the series of bivariate plots indicating that Al, Fe, clay, and organic matter content are weakly related to metal concentrations.

Local factors along the study reach may be driving the changes in metal concentrations spatially and temporally. Different sedimentation rates along the study reach could result in some near-channel positions

receiving inputs of relatively cleaner (or more contaminated) sediments between 2007 and 2019, whereas at other near-channel positions with no sedimentation the same sediment would have been sampled in 2007 and 2019. In this scenario, sites lacking inputs of new sediment would have persistently high metal concentrations and reveal little change between 2007 and 2019. Although sedimentation rate data are not

available for the Lahn River study area, I do have sedimentation rate calculations from the Dill River, a tributary to the Lahn River (Martin, 2015). There, sedimentation rates were calculated at $0.3\text{--}0.4\text{ cm yr}^{-1}$, or potentially $5.7\text{--}7.6\text{ cm}$ between 2000 and 2019. Given that the sampling depth in all three dates was 5 cm, it is possible that the 5 cm samples in 2019 were from different sediment than the 5 cm samples collected in 2000 or 2007. Field evidence that would allow me to identify such sites is lacking, but the “grab samples” noted in this study were collected from single-event flood deposits that were $\sim 1.5\text{ cm}$ thick, so $\geq 5\text{ cm}$ of sedimentation between 2000 and 2019 is reasonable. As shown in Figs. 2 and 3, temporal variation in metal concentrations (both original and normalized) among the three sample dates is greatest for Cu and Zn in the Gi-Wz river section, whereas there is little difference in the temporal variation for Pb across the three river sections. If the differences in metal concentrations for Cu and Zn in the Gi-Wz section are caused by samples being collected from different sediments in 2000, 2007, and 2019, one would expect the temporal variation for Pb to also be at a maximum in that reach. The fact that it is not suggests that different sedimentation rates do not play a major role in explaining metal concentrations among the three sample dates. Additionally, if sedimentation rates were a major factor in accounting for differences among the 2000, 2007 and 2019 samples, one would expect more consistent changes at given sites for all three metals over the three time periods. For example, if the addition of new sediment at a given site sometime between 2000 and 2019 was an important factor, then the 2019 values for Cu, Pb, and Zn should all be either greater, roughly equal to, or less than the 2000 values. In fact, for many individual sites the 2019 Pb value is clearly greater than the 2000 and 2007 Pb values, but the 2019 Cu and Zn values are equal to or less than the 2000 and 2007 values. As noted earlier, on a sample site basis, the Cu and Zn concentrations increased at 16 and 15 sites, respectively, between 2007 and 2019, whereas the Pb concentration increased at 23 sites during the same time period. If sedimentation between 2007 and 2019 was impacting changes in metal concentrations, the number of sites where the concentrations increased should be roughly equal for all the three metals. Additional inputs of Pb to the study reach between 2007 and 2019 are a more plausible explanation for Pb concentrations increasing at more sites than did Cu or Zn concentrations.

Linked to a potential spatial variation in sedimentation between 2007 and 2019 is the possibility that some parts of the study reach experienced overbank flows whereas other parts did not. In this scenario, those areas that were flooded would have received inputs of new sediments that would then have been sampled in 2019, or alternatively experienced erosion of the floodplain surface that removed the sediments sampled in 2007. Arguing against erosion of the floodplain surface is the thick vegetation cover present in a several-meter wide near-channel buffer zone on both sides of the Lahn River throughout nearly all of the study reach. In addition, overbank flow in the study reach tends to be low velocity given the gentle gradient of the river ($0.02^\circ\text{--}0.14^\circ$) throughout the study reach. Finally, numerous step-dams built in the Lahn River during the mid-1800s to facilitate navigation reduce river and overbank velocity. In many years of work along the Lahn River before, during, and after high water events, I have never observed significant near-channel soil/surface sediment erosion. Rather, vertical accretion such as documented in March 2019 and sampled at that time is far more common. Documenting the size, timing, and frequency of overbank flows in the study reach is complicated by the fact that the closest gauging station to the study reach is located in Dietz, about 35 km downstream from the end of the study reach. In addition, collection of discharge data began on 1 January 2012. Thus, the temporal and spatial resolution of the discharge data provides little information on overbank events that might have impacted the study reach between 2000 and 2019.

Nonetheless, it is possible that overbank flow events between 2007 and 2019 differentially impacted the study reach and played a role in the observed changes in metal concentrations over that period. But as was

noted above in the discussion of differences in sedimentation rates, one would expect sites that received new sediment to show a consistent difference in concentrations for all three metals when comparing 2000, 2007 and 2019 data. As previously stated, if sediment events between 2000 and 2019 were impacting changes in metal concentrations, the number of sites where the concentrations increased should be roughly equal for the three metals. In fact, they are not as detailed above. The complex spatial pattern of metal concentrations is as apparent in the 2019 data as it was in the 2000 and 2007 data. “Hotspots” with anomalously high metal concentrations are scattered through the 75-km study reach; Martin (2015) found a similar spatial pattern in near-channel locations along the nearby Dill River. The spatial patterns could be related to geomorphic processes (Dhivert et al., 2016) that operate over small spatial scales to create a variable pattern of metal deposition and storage, although that would require processes operating differently on Pb than on Cu and Zn. Differences in topography (e.g., floodplain swales vs. point bars), another factor that has been shown to impact metal storage (Lecce and Pavlowsky, 2001; Szabó et al., 2020), is another possible explanatory factor, but again that would require that topography is affecting accumulation of Cu and Zn differently than accumulation of Pb. My previous work along transects perpendicular to the Lahn River has revealed no differences in metal concentrations between swales and flatter areas of the floodplain (Martin, 2009). The different spatial variability of Cu/Zn in contrast to Pb may reflect multiple sources for the latter metal in the study area, as noted by Lecce and Pavlowsky (2014). Multiple sources for Pb have resulted in a more even distribution of the metal (that is, less site-to-site lateral and downstream variability) in the study reach.

6. Conclusion

Concentrations of Cu and Zn at near-channel positions along the Lahn River, central Germany, were unchanged between 2007 and 2019; in contrast, the concentration of Pb increased over the period. The Cu and Zn concentrations increased at 48 % and 45 %, respectively, of the 33 sample sites between 2007 and 2019. In contrast, Pb concentrations increased at 70 % of the sample sites over the study period. When the study reach is divided into three river segments, Cu, Pb, and Zn median and mean concentrations increased in the Mb-Gi and Wz-Wb sections between 2007 and 2019; in the third section (Gi-Wz), mean and median concentrations of Cu and Zn and the median concentration of Pb decreased whereas the mean concentration of Pb increased. The median concentrations of all three metals are statistically higher than background values, but only Zn concentrations exceed the 26 background upper limit. The most downstream segment of the study reach (Wz-Wb) continued to have the highest concentrations of all three trace metals.

The spatial pattern downstream of metal concentrations remains variable. Individual near-channel sites with relatively high metal concentrations, sometimes referred to as “hotspots”, continue to appear in the 2019 samples. Although I cannot identify the exact mechanism that results in such hotspots, their existence appears to be independent of factors such as differences in substrate, inherent non-anthropogenic spatial variability in metal concentrations, variable sedimentation rates and flood history, and clay and organic matter content. Rather, the existence of sites with high metal contents suggests that input of metals to the river system continues. That is, they reflect source inputs rather than other factors. Locations with higher metal concentrations also represent problematic potential sources for downstream metal pollution into the future. Indeed, they may already be serving as sources, along with former mining activity, for metal pollution noted in reaches of the Lahn River downstream from the study reach (Hahn et al., 2021).

Increasing Pb concentrations in near-channel sediments along most of the entire study reach suggests a reversal from previous years in the conveyance of Pb-contaminated sediments through the river. At least between 2007 and 2019, cleaner sediments were not being deposited in near-channel positions. It is possible that “saturation”, or the storage of

Pb in upstream soils and sediments (Nriagu et al., 1998; Outridge et al., 2011), is producing a lag between the decline in metal emissions to the environment and the decrease in metal concentrations in downstream depositional environments. Indeed, in the case of Pb throughout the study reach, and Cu and Zn in a large part of it, the reduction in metal emissions to the environment is currently not resulting in decreasing metal concentrations moving through, and being deposited along, the Lahn River. Between 2007 and 2019, near channel metal concentrations appear to have become “uncoupled” from the trend in metal releases to the environment. An increase in Cu, Pb, and Zn emissions to the environment between 2007 and 2019 is possible, but unlikely given the prevalence of strict environmental controls in Germany. It is more likely that the river, or surface water systems flowing into the river, are conveying more metal-rich sediments downstream. Surface soils, and sub-surface sediments of channel banks along the Lahn River, have been shown to contain concentrations of all three trace metals that exceed background values (Martin, 2004), with maximum metal concentrations found either at the surface or just below it (Martin, 2019). Additionally, relatively high concentrations of Pb are present in soils and surface sediments laterally across the Lahn River floodplain (Martin, 2009). Higher-than-background concentrations of Cu, Pb, and Zn are also present just below the surface at near-channel locations along the Dill River (Martin, 2015), which enters the Lahn River at Wetzlar.

A limited picture of the metal concentrations currently being conveyed through the Lahn River is provided by the three “grab” samples collected from freshly deposited flood silts. All three samples contained concentrations of Cu, Pb, and Zn that are above background concentrations, an indication that anthropogenic metals continue to be transported through the Lahn River by flood events. The results from the “grab” samples demonstrate that upstream locations remain sources for anthropogenic metal contamination in the study reach. Simply, metal concentrations equal to or less than background levels are not presently being conveyed through the river system. Rather, elevated concentrations of trace metals persist in near-channel positions of the Lahn River study reach. Reducing heavy metal concentrations stored in channel banks of the river could be a lengthy process that will lag behind the already-realized reductions in metals to the environment and may periodically involve reversals in metal trends, as appears to have happened here with Pb concentrations between 2007 and 2019.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Bindler, R., Rydberg, J., Renberg, I., 2011. Establishing natural sediment reference conditions for metals and the legacy of long-range and local pollution on lakes in Europe. *J. Paleolimnol.* 45, 519–531.

- Birch, G.F., Robertson, E., Taylor, S.E., McConchie, D.M., 2000. The use of sediments to detect human impact on the fluvial system. *Environ. Geol.* 39, 1015–1028.
- Brännvall, M.L., Bindler, R., Renberg, I., Emteryd, O., Bartnicki, J., Billström, K., 1999. The Medieval metal industry was the cradle of modern large-scale atmospheric lead pollution in northern Europe. *Environ. Sci. Technol.* 33, 4391–4395.
- Brügmann, L., 1995. Metals in sediments and suspended matter of the river Elbe. *Sci. Total Environ.* 159, 53–65.
- Buchty-Lemke, M., Hagemann, L., Maaß, A.-L., Schütttrumpf, H., Schwarzbauer, J., Lehmkuhl, F., 2019. Floodplain chronology and sedimentation rates for the past 200 years derived from trace element gradients, organic compounds, and numerical modeling. *Environ. Earth Sci.* 78, 445. <https://doi.org/10.1007/s12665-019-8428-4>.
- Ciszewski, D., Turner, J., 2009. Storage of sediment-associated heavy metals along the channelized Odra River, Poland. *Earth Surf. Process. Landf.* 34, 358–372.
- Ciszewski, D., Grygar, T.M., 2016. A review of flood-related storage and mobilization of heavy metal pollutants in river systems. *Water Air Soil Pollut.* 227, 239. <https://doi.org/10.1007/s11270-016-2934-8>.
- Clement, A.J.H., Nováková, T., Hudson-Edwards, K.A., Fuller, I.C., Macklin, M.G., Fox, E. G., Zapico, I., 2017. The environmental and geomorphological impacts of historical gold mining in the Ohinemuri and Waihou river catchments, Coromandel, New Zealand. *Geomorphology* 295, 159–175.
- Coulthard, T.J., Macklin, M.G., 2003. Modeling long-term contamination in river systems from historical metal mining. *Geology* 31, 451–454.
- Davies, B.E., 1974. Loss-on-ignition as an estimate of soil organic matter. *Soil Sci. Soc. Am. J.* 38, 150–151.
- Dennis, I.A., Coulthard, T.J., Brewer, P., Macklin, M.G., 2009. The role of floodplains in attenuating contaminated sediment fluxes in formerly mined drainage basins. *Earth Surf. Process. Landf.* 34, 453–466.
- Dhivert, E., Grosbois, C., Courtin-Nomade, A., Bourrain, X., Desmet, M., 2016. Dynamics of metallic contaminants at a basin scale – spatial and temporal reconstruction from four sediment cores (Loire fluvial system, France). *Sci. Total Environ.* 541, 1504–1515.
- Esser, V., Buchty-Lemke, M., Schulte, P., Podzum, L.S., Lehmkuhl, F., 2020. Signatures of recent pollution profiles in comparable central European rivers – examples from the international River Basin District Meuse. *Catena* 193, 104646. <https://doi.org/10.1016/j.catena.2020.104646>.
- Förstner, U., 2003. Geochemical techniques on contaminated sediments – river basin view. *Environ. Sci. Pollut. Res. Int.* 10, 58–68.
- Foulds, S.A., Brewer, P.A., Macklin, M.G., Haresign, W., Betson, R.E., Rassner, S.M.E., 2014. Flood-related contamination in catchments affected by historical metal mining: an unexpected and emerging hazard of climate change. *Sci. Total Environ.* 476 (477), 165–180.
- Geiselbrecht, A., Rouhani, S., Thorbjørnsen, K., Blue, D., Nadeau, S., Gardner-Brown, T., Brown, S., 2019. Important considerations in the derivation of background at sediment sites. *Integr. Environ. Assess. Manag.* 15 (3), 448–457.
- Grosbois, C., Meybeck, M., Horowitz, A., Ficht, A., 2006. The spatial and temporal trends of Cd, Cu, Hg, Pb, and Zn in Seine River floodplain deposits (1994–2000). *Sci. Total Environ.* 356, 22–37.
- Hahn, J., Zils, V., Berresheim, L., 2021. Joint recording of contamination status, multi-element dynamics, and source identification on a sub-catchment scale: the example Lahn River (Germany). *Sci. Total Environ.* 762, 143110. <https://doi.org/10.1016/j.scitotenv.2020.143110>.
- Ho, H.H., Swennen, R., Cappuyns, V., Vassilieva, E., Tran, T.V., 2012. Necessity of normalization to aluminum to assess the contamination by heavy metals and arsenic in sediments near Haiphong Harbor, Vietnam. *J. Asia Earth Sci.* 56, 229–239.
- Horowitz, A.J., 1991. A Primer on Trace Metal-Sediment Chemistry, 2nd edition. US Geological Survey Open-File Report 91-76, Denver, CO.
- Ilie, M., Marinescu, F., Szepe, R., Ghita, G., Deak, G., Anghel, A.-M., Petrescu, A., Urişescu, B., 2017. Ecological risk assessment of heavy metals in surface sediments from the Danube River. *Carpathian J. Earth Environ. Sci.* 12 (2), 437–445.
- James, L.A., 1989. Sustained storage and transport of hydraulic gold mining sediment in the Bear River, California. *Ann. Assoc. Am. Geogr.* 79, 570–592.
- James, L.A., 2013. Legacy sediment: definitions and processes of episodically produced anthropogenic sediment. *Anthropocene* 2, 16–26.
- Jöckel, R., 1981. Geochemische Untersuchung der Lahnsedimente im Hinblick auf Ihre Belastung durch Schwermetalle. Diplomarbeit, Fachbereich Geowissenschaften und Geographie der Justus Liebig-Universität, Gießen, Germany.
- Kincey, M., Warburton, J., Brewer, P., 2018. Contaminated sediment flux from eroding abandoned historical mines: spatial and temporal variability. *Geomorphology* 319, 199–215.
- Krüger, F., Meissner, R., Gröngroft, A., Grunewald, K., 2005. Flood induced heavy metal and arsenic contamination of Elbe River floodplain soils. *Acta Hydrochim. Hydrobiol.* 33, 455–465.
- Lecce, S., Pavlowsky, R., 1997. Storage of mining-related zinc in floodplain sediments, Blue River, Wisconsin. *Phys. Geogr.* 18, 424–439.
- Lecce, S., Pavlowsky, R., 2001. Use of mining-contaminated tracers to investigate the timing and rates of historical flood plain sedimentation. *Geomorphology* 38, 85–108.
- Lecce, S., Pavlowsky, R., Schlomer, G., 2008. Mercury contamination of active channel sediment and floodplain deposits from historic gold mining at Gold Hill, North Carolina, USA. *Environ. Geol.* 55, 113–121.
- Lecce, S.A., Pavlowsky, R.T., 2014. Floodplain storage of sediment contaminated by mercury and copper from historic gold mining at Gold Hill, North Carolina, USA. *Geomorphology* 206, 122–132.
- Le Gall, M., Ayrault, S., Evrard, O., Lacey, J.P., Gateuille, D., LeFèvre, I., Mouchel, J.-M., Meybeck, M., 2018. Investigating the metal contamination of sediment transported by the 2016 Seine River flood (Paris, France). *Environ. Pollut.* 240, 125–139. <https://doi.org/10.1016/j.envpol.2018.04.082>.

- Leigh, D.S., 1997. Mercury-tainted overbank sediment from past gold mining in North Georgia, USA. *Environ. Geol.* 30, 244–251.
- Lokas, E., Wachniew, P., Ciszewski, D., Owczarek, P., Chau, N.D., 2010. Simultaneous use of trace elements, ²¹⁰Pb and ¹³⁷Cs in floodplain sediments of a lowland river as indicators of anthropogenic impacts. *Water Air Soil Pollut.* 207, 57–71.
- Loring, D.H., 1991. Normalization of heavy-metal data from estuarine and coastal sediments. *ICES J. Mar. Sci.* 48, 101–115.
- MacDonald, D.D., Ingersoll, T.A., Berger, T.A., 2000. Development and evaluation of consensus-based sediment quality guidelines for freshwater ecosystems. *Arch. Environ. Contam. Toxicol.* 39, 20–31.
- Macklin, M.G., Brewer, P.A., Hudson-Edwards, K.A., Bird, G., Coulthard, T.J., Dennis, I. A., Lechler, P.J., Miller, J.R., Turner, J.N., 2006. A geomorphological approach to the management of river contaminated with metal mining. *Geomorphology* 79, 423–447.
- Marcus, W.A., 1989. Regulating contaminated sediments in aquatic environments: a hydrologic perspective. *Environ. Manag.* 13, 703–713.
- Martin, C.W., 2004. Heavy metal storage in near channel sediments of the Lahn River, Germany. *Geomorphology* 61, 275–285.
- Martin, C.W., 2009. Recent changes in heavy metal storage in flood-plain soils of the Lahn River, Central Germany. *Environ. Geol.* 58, 803–814.
- Martin, C.W., 2012. Recent changes in heavy metal contamination at near-channel positions of the Lahn River, Central Germany. *Geomorphology* 139 (149), 452–459.
- Martin, C.W., 2015. Trace metal storage in recent floodplain sediments along the Dill River, Central Germany. *Geomorphology* 235, 52–62.
- Martin, C.W., 2019. Trace metal concentrations along tributary streams of historically mined areas, lower Lahn and Dill River basins, Germany. *Catena* 174, 174–183. <https://doi.org/10.1016/j.catena.2018.11.008>.
- Matache, M.L., David, I.G., Dinu, C., Radu, L.G., 2018. Trace metals in water and sediments of the Prut River, Romania. *Environ. Eng. Manag. J.* 17 (6), 1363–1371.
- Matschullat, J., Ellminger, F., Agdemir, N., Cramer, S., Ließmann, W., Nichoff, N., 1997. Overbank sediment profiles – evidence of early mining and smelting activities in the Harz Mountains, Germany. *Appl. Geochem.* 12, 105–114.
- Matschullat, J., Ottenstein, R., Reimann, C., 2000. Geochemical background – can we calculate it? *Environ. Geol.* 39, 990–1000.
- Meybeck, M., Lestel, L., Bonté, P., Moilleron, R., Colin, J.L., Rousselot, O., Hervé, D., de Pontevès, C., Grosbois, C., Thévenot, D.R., 2007. Historical perspective of heavy metals contamination (Cd, Cr, Cu, Hg, Pb, Zn) in the Seine River basin (France) following a DPSIR approach (1950–2005). *Sci. Total Environ.* 375, 204–231.
- Meyer, F., 2007. Die Geschichte des Bergbaus im Weilburger Land. Libelli: Museum Extra Nr. 1. Bergbau- und Stadtmuseum Weilburg, Weilburg an der Lahn, Germany.
- Middelkoop, H., 2002. Reconstructing floodplain sedimentation rates from heavy metal profiles by reverse modeling. *Hydrol. Process.* 16, 47–64.
- Murad, E., Fischer, W.R., 1978. Mineralogy and heavy metal contents of soils and stream sediments in a rural region of western Germany. *Geoderma* 21, 133–145.
- Nriagu, J.O., Wong, H.K.T., Lawson, G., Daniel, P., 1998. Saturation of ecosystems with toxic metals in Sudbury basin, Ontario, Canada. *Sci. Total Environ.* 223, 99–117.
- Outridge, P.M., Rausch, N., Percival, J.B., Shotyk, W., McNeely, R., 2011. Comparison of mercury and zinc profiles in peat and lake sediment archives with historical changes in emissions from the Flin Flon metal smelter, Manitoba, Canada. *Sci. Total Environ.* 409, 548–563.
- Owens, P.N., Batalla, R.J., Collings, A.J., Gomez, B., Hicks, D.M., Horowitz, A.J., Kondolf, G.M., Marden, M., Page, M.J., Peacock, D.H., Petticrew, E.L., Salomons, W., Trustrum, N.A., 2005. Fine-grained sediment in river systems: environmental significance and management issues. *River Res. Appl.* 21, 693–717.
- Renberg, I., Bindler, R., Brännvall, M.L., 2001. Using the historical atmospheric lead-deposition record as a chronological marker in sediment deposits in Europe. *The Holocene* 11, 511–516.
- Resongles, E., Casiot, C., Freyrier, R., Dezileau, L., Viers, J., Elbaz-Poulichet, F., 2014. Persisting impact of historical mining activity to metal (Pb, Zn, Cd, Tl, Hg) and Metalloid (As, Sb) enrichment in sediments of the Gardon River, southern France. *Sci. Total Environ.* 481, 509–521.
- Salomons, W., Förstner, U., 1984. *Metals in the Hydrocycle*. Springer-Verlag, Berlin.
- Scherer, U., Fuchs, S., Behrendt, H., Hillenbrand, T., 2003. Emissions of heavy metals into river basins of Germany. *Water Sci. Technol.* 47, 251–257.
- Schwartz, R., Gerth, J., Neumann-Hensel, H., Bley, S., Förstner, U., 2006. Assessment of highly polluted Fluviosol in the Spittelwasser floodplain. *J. Soils Sediments* 6, 145–155.
- Schropp, S.J., Lewis, F.G., Windom, H.L., Ryan, J.D., Calder, F.D., Burney, L.C., 1990. Interpretation of metal concentration in estuarine sediments of Florida using aluminum as a reference element. *Estuaries* 13 (3), 227–235.
- Szabó, Z., Buró, B., Szabó, J., Tóth, C.A., Baranyai, E., Herman, P., Prokisch, J., Tomor, T., Szabó, S., 2020. Geomorphology as a driver of Heavy metal accumulation patterns in a floodplain. *Water* 12, 563. <https://doi.org/10.3390/w12020563>.
- Štrbac, S., Grubin, M.K., Vasić, N., 2018. Importance of background values in assessing the impact of heavy metals in river ecosystems: case study of Tisza River, Serbia. *Environ. Geochem. Health* 40, 1247–1263. <https://doi.org/10.1007/s10653-017-0053-0>.
- Wadige, C.P.M.M., Taylor, A.M., Krikowa, F., Maher, W.A., 2016. Sediment metal concentration survey along the mine-affected Molonglo River, NSW, Australia. *Arch. Environ. Contam. Toxicol.* 70, 572–582.
- Wang, Z., Hua, P., Ruifei, L., Bai, Y., Fan, G., Wang, P., Hu, B.X., Zhang, J., Krebs, P., 2019. Concentration decline in response to source shift of trace metals in Elbe River, Germany: a long-term trend analysis during 1998–2016. *Environ. Pollut.* 250, 511–519.
- Zerling, L., Hanische, C., Junga, W., 2006. Heavy metal inflow into the floodplains at the mouth of the river Weiße Elster (central Germany). *Acta Hydrochim. Hydrobiol.* 34, 234–244.